KANSAS UNDERGROUND INJECTION CONTROL PERMIT
CLASS V EMPLACEMENT WELLS

Pursuant to the provisions of Kansas Statutes Annotated (K.S.A.) 65-164, 65-165, 65-166, 65-170g and 65-171d and Kansas Administrative Regulations (K.A.R.) Chapter 28, Article 46, the undersigned is authorized to emplace materials that meets the approved beneficial reuse material acceptance criteria found within this permit. The emplacement shall be done in accordance with the construction, operation, monitoring, and reporting requirements as set forth herein.

Owner:	Underground Cavern Stabilization, LLC Post Office Box 225 Great Bend, KS 67530
Operator:	Advantek Cavern Solutions, LLC 11000 Richmond Ave Suite 190 Houston, TX 77042
Facility:	Advantek Cavern Solutions 7513 South K-14 Highway Hutchinson, KS 67501
Facility Contact:	Steve Pangburn - Facility Director Jay Cecil – VP of Operations
Facility Telephone Number:	620.662.6367 713.532.7627
Well Location and Identification:	See Table 1
Receiving Formation:	Cavern Solutioned Mined in the Hutchinson Salt

The permittee shall comply with all conditions in this permit, federal and state regulations governing Class V injection wells and the requirements of the Kansas Department of Health and Environment (KDHE). This permit is effective June 5th, 2024 and will expire June 5th, 2034.

Janet Stanek

Janet Stanek
Secretary
Kansas Department of Health and Environment (KDHE)

6-10-2024
Date

FACILITY DESCRIPTION

Advantek Cavern Solutions, LLC (ACS) is authorized to emplace materials that meets the approved beneficial reuse material acceptance criteria found in Attachment II and Attachment III of this permit. These materials may include chloride, salt, or mineral impacted soils and mineral based material, crude oil impacted soil from spills and cleanups resulting from the oil and gas production industry, drill cuttings from exploration activity, and biosolids and semi-biosolids from animal manures, agricultural residuals and plant wastes, and other organic sources. The emplacement is for the beneficial purpose of stabilization of salt caverns solution mined in the Hutchinson Salt and removal of the material from the environment. No other materials will be allowed without prior approval from KDHE.

Table 1: List of wells at the ACS Facility.

Well ID	Latitude	Longitude	Status
AM-1	37.969156	-97.950466	Plugged
AM-2	37.9691	-97.95158	Plugged
AM-3	37.968343	-97.950434	Plugged
AM-4	37.968326	-97.951403	Plugged
AM-5	37.967525	-97.950421	Plugged
AM-6	37.96761	-97.95159	Plugged
AM-7	37.96673	-97.95045	Plugged
AM-8	37.96664	-97.95174	Plugged
AM-9	37.9691	-97.95248	Plugged
AM-10	37.968328	-97.952527	Plugged
AM-11	37.96738	-97.95265	Plugged
AM-12	37.962498	-97.949345	Plugged
AM-14	37.96908	-97.95358	Plugged
AM-15	37.96765	-97.9517	Plugged
AM-16	37.96738	-97.95369	Plugged
AM-17	37.96646	-97.95378	Plugged
AM-18	37.96601	-97.95042	Plugged
AM-19	37.965157	-97.95157	Plugged
AM-20	37.966	-97.95281	Plugged
AM-21	37.96599	-97.95367	Plugged
AR-1	37.968938	-97.94835	Plugged
AR-2	37.968525	-97.949059	Plugged
AR-3	37.968084	-97.948361	Plugged
AR-4	37.967681	-97.949039	Reopened
AR-5	37.967289	-97.948375	Reopened
AR-6	37.969312	-97.947691	Plugged

Well ID	Latitude	Longitude	Status
AR-7	37.968517	-97.947655	Plugged
AR-8	37.967689	-97.947678	Plugged
AR-9	37.966877	-97.947646	Plugged
AR-10	37.968908	-97.946946	Plugged
AR-11	37.96812	-97.946953	Plugged
SW-11	37.966	-97.94909	Plugged
SW-12	37.96601	-97.94781	Active Emplacement
SW-13	37.96603	-97.94652	Plugged
SW-14	37.96604	-97.94524	Plugged
SW-15	37.96605	-97.94395	Plugged
SW-16	37.96606	-97.94267	Plugged
SW-17	37.96607	-97.94139	Plugged
SW-21	37.96538	-97.94844	Plugged
SW-22	37.96539	-97.94716	Plugged
SW-23	37.9654	-97.94587	Plugged
SW-24	37.96541	-97.9446	Plugged
SW-25	37.96542	-97.94331	Plugged
SW-31	37.96474	-97.94908	Plugged
SW-32	37.96475	-97.94781	Plugged
SW-34	37-96478	-97.94525	Plugged
SW-41	37.96412	-97.94844	Plugged
SW-42	37.96413	-97.94717	Plugged
SW-43	37.96414	-97.94587	Plugged
SW-44	37.96415	-97.94459	Plugged
SW-45	37.96416	-97.9433	Plugged
SW-51	37.96348	-97.94908	Plugged
SW-52	37.9635	-97.9478	Plugged
SW-53	37.96351	-97.94651	Plugged
SW-54	37.96352	-97.94523	Plugged
SW-55	37.96353	-97.94395	Plugged
SW-61	37.96286	-97.94844	Plugged
SW-62	37.96287	-97.94715	Plugged
SW-63	37.96288	-97.94587	Plugged
SW-64	37.96287	-97.94459	Plugged
SW-65	37.9629	-97.9433	Plugged

FACILITY LOCATION MAP

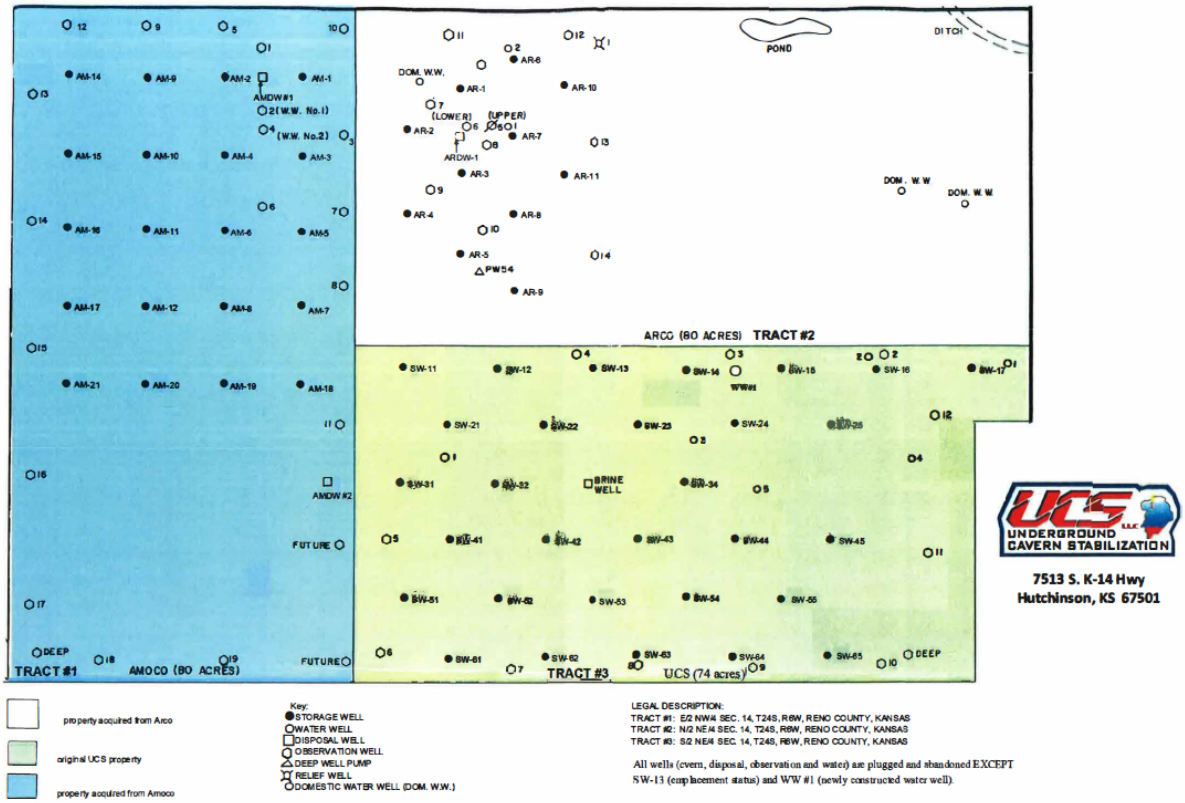


Figure 1 | ACS Facility

SECTION I CONSTRUCTION REQUIREMENTS

A. General Construction Requirements: The permittee shall design, construct, maintain, and operate each permitted well to prevent the possibility of emplaced fluids being introduced into a useable aquifer or any underground source of drinking water ensuring that emplaced fluids or other formation fluids do not cause deterioration of the water quality of fresh and/or usable water zones. The well shall be cased and cemented such that it must prevent the loss of fresh and/or usable water due to downward migration and constructed as to prohibit the release of emplaced fluids into an unauthorized zone. Construction shall also include adherence to operating conditions and procedures, and emergency shutdown procedures specified in the permit application and in this permit.

B. Emplacement Zone: Hutchinson Salt through open-hole from:

Well Identification	Open-Hole Intervals
SW-12	530 feet to 755 feet
AR-04	562 feet to 778 feet
AR-05	564 feet to 781 feet

C. Emplacement Pressure: Emplacement pressure shall be continuously monitored and shall not exceed a wellhead pressure of 80 psig and a pressure gradient of 0.75 psi/ft as calculated at the cavern roof. If the emplacement pressure exceeds 80 psig, the permittee shall notify KDHE as specified in Attachment I, paragraph O.

D. Spill Prevention and Containment: The processing equipment and emplacement wellhead are located within a bermed containment structure lined with HDPE geomembrane. The containment structure shall be kept essentially empty at all times. The facility shall keep an updated Spill Response Plan onsite and shall make the plan available to KDHE upon request.

Table 1: Borehole, casing, tubing, and cement specifications for each well in this project:

Well ID		Borehole Size (in)	Casing or Tubing Size (in) & Material	Weight (lbs/ft)	Casing Seat Depth (ft)	Type of Cement & Additives	Number of Sacks of Cement
SW-12		Unknown	8-5/8" Steel	24	526'	Unknown	Unknown
		NA	5-1/2" Steel				
AR-04		Unknown	9-5/8"	Unknown	562'	Unknown	Unknown
		NA					
AR-05		Unknown	9-5/8"	Unknown	564'	Unknown	Unknown
		NA					

NA = Not Applicable

SECTION II EMPLACEMENT LIMITATIONS, MONITORING, REPORTING AND TESTING REQUIREMENTS

A. General Operations: Approved materials will be mixed with saturated brine and injected through the inner tubing string into the open hole cavern space; brine from within the cavern will be displaced and circulated to the surface between the inner tubing string and casing. Approved bio-oil materials will be mixed with salt before injection to saturate the water content of the bio-oil; the bio-oil material will then be injected through the inner tubing string into the open hole cavern space and brine from within the cavern will be displaced and circulated to the surface between the inner tubing string and casing. The displaced brine will be contained in above ground tanks for future use or injected into the Arbuckle Formation under the facility's UIC Class I permit.

B. Monitoring Data: Monthly monitoring data required to be submitted to KDHE via the Kansas Online Automated Reporting (KOLAR) website monthly, shall be submitted no later than 28 days after the last day of the month for which the monitoring data are being reported. Quarterly monitoring data required to be submitted to KDHE via the KOLAR website on a quarterly basis shall be submitted no later than 28 days after the last day of the calendar quarter for which monitoring data are being reported. All monitoring data required for reports shall be submitted to the KOLAR website hosted by the Kansas Geological Survey.

C. Permit Limits and Inspection Reporting: Inspection readings of emplacement fluid flow rate, volume, wellhead emplacement pressure, displacement flow rate, volume, and displacement wellhead pressure shall be made daily and reported in the monthly monitoring report submitted to KDHE via KOLAR. The inspection readings for the wellhead pressures and fluid flow rates shall include readings of both the gauge or meter and the continuous recording devices. The inspection reading of the fluid volumes shall be a reading of the meter. The date and time these readings are taken and the initials of the person taking the readings shall be included in the monthly monitoring report.

The monthly average, maximum and minimum values for the daily inspection readings of flow rates, volumes, and wellhead pressures for the month shall be reported in the monthly monitoring report submitted to KDHE via KOLAR. The monthly average flow rates and volumes shall be calculated using only the number of days for which emplacement occurred during the month. The total volume of fluid emplaced and displaced for the month shall also be reported in the monthly monitoring report submitted to KDHE via KOLAR.

The maximum and minimum values determined from the continuous recording data for the entire month for wellhead pressures shall be reported in the monthly monitoring report submitted to KDHE via KOLAR.

Table 2: Permittee Monitoring Schedule: Measure Daily, Report Monthly

Emplacement and Displacement Operational Parameters	Emplacement or Parameter Limitation	Sample or Measurement Type
Maximum Pressure	80 psig wellhead-pressure and 0.75 psi/ft. pressure gradient as calculated at the cavern top	Gauge and Electronic Continuous Recording Device*
Maximum Daily Emplacement Volume	12,000 bpd	Meter and Electronic Continuous Recording Device*
Maximum Rate	Monitor	Meter and Electronic Continuous Recording Device*

* The gauge, meter, detector, and electronic continuous recording device shall at all times be maintained operational and shall at all times be located to properly measure the activity being monitored.

D. Analytical Samples and Reporting: The analytical parameters and required reporting for emplacement and displacement fluids are listed in Table 3 through Table 6.

1. Bio-Oil Material Emplacement: the permittee shall sample the emplacement material and attach documentation for each day that bio-oil material is being emplaced and shall sample the displaced brine each day emplacement occurs. The documentation shall include the water fraction of the material and analytics that prove that the water fraction of the emplaced bio-oil has been saturated to a chloride concentration of at least 190,000

mg/L as specified in Table 3. The bio-oil material must be handled and mixed in accordance with SCS Engineers' Standard Operating Procedures (Attachment VI). The documentation shall be attached to the monthly monitoring report and submitted to KDHE via the KOLAR website monthly, no later than 28 days after the last day of the month for which the data is being reported.

Table 3: Bio-Oil Material Emplacement and Displacement Requirements and Sampling

Emplacement Attach documentation for each day that bio-oil material is emplaced, report monthly	
Fluid density (ppg)	Monitor
Conductivity (Mmhos/cm)	Monitor
Temperature (°F)	Monitor
Chloride (mg/L) concentration of the water fraction of the bio-oil material	At least 190,000
pH	3.0-11.0

2. Organic Material Emplacement: the permittee shall sample the emplacement slurry and the displaced brine each day that organic material is being emplaced and report each as listed in Table 4. Samples shall be collected from the emplacement slurry during emplacement from the outflow line of the mix/slurry tank leading to the emplacement wellhead. Sample results shall be submitted to KDHE via the KOLAR website monthly, no later than 28 days after the last day of the month for which the sample data is being reported. All emplacement data that is reported shall include the date the sample was collected, the date the sample was analyzed, analytical results, the name of the laboratory conducting the analyses and the laboratory certification number. Analyses of emplacement fluid required by Table 4 shall be conducted by a laboratory certified by KDHE to analyze for the constituents listed in Table 4.

Table 4: Organic Material Emplacement and Displacement Requirements and Sampling

Emplacement Grab sample daily, report monthly	
Fluid density (ppg)	Monitor
Conductivity (Mmhos/cm)	Monitor
Temperature (°F)	Monitor
Chloride (mg/L)	At least 190,000
pH	3.0-11.0

Emplacement Grab Sample Monthly, Report Quarterly	
Organic content %	Monitor
Total Nitrogen (mg/l)	Monitor
Total Phosphorous (mg/l)	Monitor
Potassium (mg/l)	Monitor
Carbon (mg/l)	Monitor
Iron (mg/l)	Monitor
Zinc (mg/l)	Monitor
Nitrate (mg/l)	Monitor
Sulfur (mg/l)	Monitor
Geometric mean fecal coliform density (MPN/g) of total solids	Less than 2,000,000

3. All Other Approved Material Emplacement: the permittee shall sample the emplacement slurry and the displaced brine each day that all other approved material is being emplaced, and report each as listed in Table 5. Samples shall be collected from the emplacement slurry during emplacement from the outflow line of the mix/slurry tank leading to the emplacement wellhead. Sample results shall be submitted to KDHE via the KOLAR website monthly, no later than 28 days after the last day of the month for which the sample data is being reported. All emplacement data that is reported shall include the date the sample was collected, the date the sample was analyzed, analytical results, the name of the laboratory conducting the analyses and the laboratory certification number. Analyses of emplacement fluid required by Table 4 shall be conducted by a laboratory certified by KDHE to analyze for the constituents listed in Table 4.

Table 5: All Other Approved Material Emplacement Sampling

Emplacement Grab sample each day that all other approved material is being emplaced, report monthly	
Fluid density (ppg)	Monitor
Conductivity (Mmhos/cm)	Monitor
Temperature (°F)	Monitor
Chloride (mg/L)	At least 190,000
pH	3.0-11.0

4. Displacement For All Materials: the permittee shall sample the displaced brine each day that any approved material is being emplaced, and report each as listed in Table 6. The sample(s) of displaced brine shall be analyzed for chloride concentration and pH for each day that material is displaced, and the results reported in the monthly monitoring report submitted to KDHE. The sample(s) shall be a grab sample collected from the displacement brine during displacement. The data reported shall include the date the sample was collected, the date the sample was analyzed, analytical results, and the type of measurement device. Chloride concentration and pH may be measured using KDHE approved field-testing equipment, provided this equipment is calibrated in accordance with the manufacturer's recommendations and records of the calibration are made available to KDHE upon request.

Table 6: Displacement Sampling

Displacement Grab sample each day that approved materials are being emplaced, report monthly	
Chloride (mg/L)	At least 190,000
pH	3.0-11.0

E. Quarterly Reporting: The level of the emplaced material in the cavern shall be measured at least quarterly and reported to KDHE via KOLAR no later than 28 days after the quarter being reported. The submittal shall be in a format that includes a comparison between the most recent measurement and previous level measurements obtained since the effective date of the permit. Quarterly analysis, measurement and reporting frequency is based on calendar quarters.

F. Groundwater Monitoring: A quarterly groundwater monitoring report is required, which follows the quality assurance plan agreed upon with KDHE. Results of the quarterly groundwater monitoring report shall be submitted to KDHE within 60 days of obtaining the results.

G. Elevation Survey: The surface elevation of all the cavern wellheads in the project area shall be measured biannually and the results submitted to KDHE within 60 days of completing the measurement.

H. Sonar Survey: A sonar survey of the cavern shall be conducted prior to commencing the emplacement operation. Additional sonar surveys shall be conducted when the cavern is estimated to be 50, percent filled as determined using a method approved by KDHE. The results of the sonar survey and an interpretation by a person with the technical expertise to make such interpretation shall be submitted to KDHE within 45 days of completing the sonar survey. Additional sonar surveys shall be completed as requested by KDHE.

I. Combustible Gas Detector/Transmitter: A combustible gas detector/transmitter shall be located on the above ground displaced brine tank vent. A verified gas detection shall be reported to KDHE within 2 hours of discovery.

J. Other Reporting: The following shall also be reported to KDHE by the permittee:

1. A well treatment plan or workover plan shall be submitted to KDHE for review and approval prior to commencing a well treatment or workover. No well treatment or workover shall commence until the permittee has obtained approval for the well treatment or workover plan from KDHE. Any well treatment procedures used, including those associated with normal maintenance and malfunction correction, and all well workovers shall be reported to KDHE within 30 days of completion.

2. Notification to KDHE within 2 hours of discovery of all spills associated with the operation of the emplacement well or well system. Spills that do not have to be reported to the state are spills within containment structures that do not threaten public health, safety or the environment and that do not impact the soils or waters of the state.

3. Notification to KDHE of any well malfunction or failure within 24 hours of becoming aware of the circumstances.

4. The results and interpretation of mechanical integrity tests and any other tests or logs of the emplacement well or injection zone within 30 days of completion.

5. A written description and explanation of any noncompliance with the operating limitations as specified by this permit for wellhead pressure, flow volume or injection limits occurring during the month being reported shall be submitted with the monthly monitoring report.

6. When the permittee becomes aware that it failed to submit any relevant facts in a permit application or submitted incorrect information in a permit application or in any report to KDHE, the permittee shall submit such facts or corrected information to KDHE within 5 days of becoming aware of the circumstances.

7. Within 10 business days of completing the filling of a cavern, the permittee shall notify KDHE in writing that the cavern filling is complete.

K. Mechanical Integrity Test (MIT):

1. Upon reentering a previously plugged storage well and cavern, a brine hydrostatic test shall be completed. The brine hydrostatic test is designed to evaluate the internal mechanical integrity of the casing including potential leakage and the external mechanical integrity of the cavern by evaluating any significant fluid movement through vertical channels adjacent to the wellbore. Upon successful completion of at least one brine hydrostatic MIT, the permittee may continue to use a brine hydrostatic MIT, a nitrogen-brine MIT, or may request to use an alternate MIT method approved by KDHE.

2. A MIT shall be completed at least once every five years on each active emplacement well and cavern and shall be completed following reentry of any plugged storage well and cavern before active emplacement operations begin. The MIT is designed to evaluate the mechanical integrity of the casing including significant fluid movement through vertical channels adjacent to the wellbore and the mechanical integrity of the cavern. If KDHE determines that use of any well constitutes a threat to public health, safety, or the fresh and/or usable waters or soils of the state, or the release of emplaced fluid or displaced brine into any unauthorized zone may be occurring, the permittee shall immediately cease

emplacement and conduct a MIT. If determined necessary by KDHE, a MIT shall be completed when there has been a well workover.

3. The permittee shall submit a MIT plan to KDHE for review and consideration for approval at least 30 days prior to conducting any MIT. No MIT work shall commence until approval of the MIT plan has been obtained from KDHE. The casing MIT shall be witnessed by KDHE. If the well fails any MIT, the requirements of Section II, K. 2. of this permit shall be implemented by the permittee. The results and interpretation of each MIT shall be submitted to KDHE within 30 days of test completion.

L. Loss of Mechanical Integrity, Anomalous Operational Data

1. If a loss of mechanical integrity or anomalous operational data indicating a loss of mechanical integrity occurs, the permittee shall immediately investigate and identify the cause of the loss of mechanical integrity or anomalous operational data and notify KDHE within 24 hours of becoming aware of the circumstances. The results of this investigation shall be reported to KDHE within 24 hours of completion. If the well appears to be lacking mechanical integrity, the permittee shall:

   a) Immediately cease emplacement of fluids.

   b) Take all steps required by KDHE to determine the presence or absence of mechanical integrity. If the well is determined to have mechanical integrity, injection may resume after the permittee has obtained authorization from KDHE.

2. If a loss of mechanical integrity is determined pursuant to Section II, K. 1. of this permit or as the result of a MIT, the permittee shall:

   a) Immediately cease emplacement of fluids.

   b) Notify KDHE within 24 hours of the determination.

   c) Take all steps determined necessary by KDHE to determine whether there may have been a release of fluids into any unauthorized zone. If there is evidence there may have been a release into an unauthorized zone, the permittee shall orally notify KDHE within 24 hours of the determination. A written notice shall also be provided to KDHE within 5 days of the determination including a description of the release.

   d) Comply with any immediate corrective or remedial action specified by KDHE. If determined necessary by KDHE, the permittee shall submit to KDHE a remediation and corrective action plan and implementation schedule for review and approval. Work shall not commence until approval of the remediation and corrective action plan has been obtained from KDHE.

   e) Restore and demonstrate mechanical integrity to the satisfaction of KDHE. A plan for any well workover or MIT shall be submitted to KDHE for review and approval. Work shall not commence until the permittee has obtained approval of the workover or MIT plan from KDHE.

   f) Emplacement of material shall resume only upon authorization from KDHE.

SECTION III BENEFICIAL REUSE MATERIAL ACCEPTANCE PLAN

Material shall only be allowed into this facility and emplaced in a salt cavern if done so in accordance with the Beneficial Reuse Material Acceptance Plans found in Attachment II and Attachment III. The permittee shall update the Beneficial Reuse Material Acceptance Plans annually. Any changes to the Beneficial Reuse Material Acceptance Plan must be approved by KDHE before implementation. All records required by this plan shall be made available to KDHE upon request.

SECTION IV CONVERSION

A notice of conversion of the well to a use other than material emplacement as authorized by this permit shall be submitted to KDHE at least 60 days prior to conversion. A conversion plan shall be submitted with the notice to KDHE for review and approval. The well shall not begin the conversion process until approval of the conversion plan has been obtained from KDHE and any required permits are issued.

SECTION V POST CAVERN FILLING MONITORING STATUS AND ABANDONMENT PLAN

A. Monitoring Status: The permittee currently has a monitoring status plan on file with KDHE. The permittee shall revise and update the monitoring status plan when required by KDHE. Within 30 days of completing the filling of the cavern, the permittee shall submit a revised and updated post cavern fill monitoring status plan to KDHE for review and approval. The plan shall be implemented within 30 days of KDHE approval of the monitoring status plan.

B. Plugging and Abandonment: Upon meeting the requirements of the monitoring status plan, the well shall be plugged and abandoned or when determined necessary by KDHE to protect public health or the fresh and/or usable waters or the soils of the State. The permittee currently has a plugging and abandonment plan on file with KDHE.

1. The permittee shall notify KDHE at least 60 days prior to plugging and abandonment of the well. With the notice, the permittee shall submit a revised and updated plugging and abandonment plan to KDHE for review and approval. The permittee shall conform to all plugging and abandonment requirements of state and federal regulations and KDHE.

2. The well shall be plugged in a manner which will not allow the movement of fluids into or between sources of fresh and/or usable water or allow the movement of emplaced fluids out of the emplacement zone.

3. Plugging and abandonment work shall not commence until approval of the plugging and abandonment plan has been obtained from KDHE. The report of plugging and abandonment and related information shall be submitted to KDHE within 30 days after completion of the plugging operation on forms provided by KDHE.

SECTION VI FINANCIAL RESPONSIBILITY FOR PLUGGING AND ABANDONMENT

The permittee shall maintain financial responsibility and financial resources to close, plug and abandon the emplacement wells and appurtenances and for the closure of the Clear Span Storage Building including the removal and disposal of any stored material in a manner required by KDHE. The permittee shall show evidence annually of financial responsibility to KDHE by the submission of a surety bond or other adequate financial assurance such as financial statements or other materials acceptable to KDHE. Financial assurance documents and closure estimates shall be revised and updated as required by KDHE. The permittee must notify KDHE by certified mail of the commencement of voluntary or involuntary proceedings under Title 11 (Bankruptcy), U.S. Code, naming the owner or operator as debtor, within ten (10) business days after the commencement of the proceeding. A guarantor of a corporate guarantee must make such notification if named as debtor, as required under the terms of the guarantee.

SECTION VII SCHEDULE OF COMPLIANCE AND STANDARD CONDITIONS

Each well and cavern that is reentered shall not be operated until a successful casing MIT has been conducted and deemed satisfactory by KDHE. Each well and cavern that is reentered shall not be operated until all cavern re-entry drilling, casing and cementing records and all log and log interpretations have been submitted to and approved by KDHE. Upon successful completion of a MIT, this permit will be revised to accurately detail well construction and injection interval information for any reentered well and cavern. This modification will not require a public notice.

In addition to the specified conditions stated herein, the permittee shall comply with the provisions of Attachment I.

SECTION VIII FACILITY SUBMISSION SUMMARY

The following is a summary of the required facility submissions/reporting pursuant to this permit.

SUBMISSION REQUIREMENTS	DUE DATE	PERMIT CONDITION
Construction and Injection Details	Within 30 days after initial MIT has been completed	VII and Attachment I.L.
Well Malfunction or Failure	Contact within 24 hrs	II.I.3.
Spills	Contact within 2 hours of discovery	II.I.2.
Testing Malfunction or Other Emergency Reporting	Contact within 24 hrs, written contact within 5 days	II.K.1. II.K.2.b.
Monthly Monitoring	28th day of each month via email or hard copy	II.A. II.B.
Treatment or Workover Reporting	Within 30 days after completion	II.I.1.
Permit Renewal	At least 180 days prior to permit expiration date	Attachment I.B.

Permit Transfer Request	At least 30 days prior to proposed transfer	Attachment I.N.
Plugging and Abandonment Plan	At least 60 days prior to proposed plugging and abandonment	V.B.1.
Plugging and Abandonment Report	Within 30 days after completion of plugging operations	V.B.3.
Financial Assurance	As requested by KDHE	VI
Any Corrections to Submittals	Within 5 days of becoming aware of the circumstance	II.I.6.
Groundwater Monitoring Report	Within 60 days of obtaining results	II.E.
Elevation Survey	Within 60 days of completing the measurement	II.F.
Sonar Survey	Within 45 days of completing the survey	II.G.
Verified Gas Detection	Within 2 hours of discovery	II.H.
Filling of a Cavern	Written contact within 10 business days	II.I.7.
Emplacement Stream Changes	180 days prior to proposed changes	Attachment I.S.
Beneficial Reuse Material Acceptance Plan	As requested by KDHE	III.
Post Cavern Filling Monitoring Status Plan	As requested by KDHE	V
MIT Plan	At least 30 days prior to test date	II.K.2.
MIT Report	Within 30 days after test completion	II.K.2

ATTACHMENT I Effective August 4, 2021

STANDARD CONDITIONS FOR UNDERGROUND INJECTION CONTROL PERMITS

CLASS V EMPLACEMENT WELLS

A. Duty to Comply: The permittee shall comply with all conditions of the permit, and Federal and State laws and regulations. Any permit noncompliance constitutes a violation of the appropriate act or regulations and is grounds for enforcement action for permit termination, revocation and reissuance, modification, or denial of a permit renewal application.

B. Duty to Reapply: If the permittee wishes to continue an activity regulated by this permit after the expiration date of this permit, the permittee must apply for and obtain a new permit. An application to renew this permit shall be filed with KDHE at least 180 days prior to its expiration date.

C. Duty to Halt or Reduce Activity: It shall not be an acceptable defense for a permittee, in an enforcement action, that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this permit.

D. Duty to Mitigate: The permittee shall take all reasonable steps to minimize or correct any adverse impact on the environment resulting from noncompliance with this permit.

E. Proper Operation and Maintenance: The permittee shall at all times properly operate and maintain all facilities and systems of monitoring, treatment and control (and related appurtenances) which are installed or used by the permittee to achieve compliance with the conditions of this permit. Proper operation and maintenance includes effective performance, adequate funding, adequate operator staffing and training, and adequate laboratory and process controls, including appropriate quality assurance procedures. This provision requires the operation of back-up or auxiliary facilities or similar systems when necessary to maintain compliance with the conditions of the permit.

F. Permit Modifications and Terminations: After notice and opportunity for a hearing, this permit may be modified, suspended or revoked, or terminated in whole or in part during its term for cause as provided, but not limited to those set forth in K.A.R. 28-46-15 and K.A.R. 28-46-16 or if KDHE or Environmental Protection Agency standards or regulations on which the permit was based have been changed by promulgation of new or amended codes, statutes, regulations or standards or by judicial decision after the permit was issued. The permittee shall furnish to KDHE, within a reasonable amount of time, any information which KDHE may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this permit or to determine compliance with this permit. The permittee shall also furnish, upon request, copies of all records required to be kept by this permit.

G. Property Rights: This permit does not convey any property rights of any sort, or any excusive privilege.

H. Duty to Provide Information: The permittee shall furnish to KDHE, within a reasonable time, any information which KDHE may request to determine whether cause exists for modifying,

revoking, reissuing or terminating the permit, or to determine compliance with this permit. The permittee shall also furnish to KDHE, upon request, copies of reports and information required to be kept by this permit.

I. Inspection and Entry: The permittee shall allow the Secretary, or any authorized representative, upon the presentation of credentials and other documents as may be required by law, to:

1. Enter upon the permittee's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this permit;

2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this permit;

3. Inspect any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this permit; and

4. Sample or monitor for the purpose of assuring permit compliance or as otherwise authorized by the appropriate Act, any substances or parameters at any location.

J. Samples, Measurements and Records:

1. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity.

2. The permittee shall retain records of all monitoring information, including all calibration and maintenance records and all electronic data collected at a minimum of every 30 seconds for continuous monitoring instrumentation, copies of all reports required by this permit, and records of all data used to complete the application for this permit, for a period of at least three (3) years for the date of sample, measurement, report or application. This period may be extended by request of KDHE at any time and is automatically extended during the course of any unresolved enforcement action regarding this Facility.

3. The permittee shall retain records concerning the nature and composition of all emplaced fluids until three (3) years after the completion of any plugging and abandonment procedures. KDHE may require the owner or operator to deliver the records to KDHE at the conclusion of the retention period.

4. Records of monitoring information shall include:
   a) The date, exact place, and time of sampling or measurements;
   b) The individual(s) who performed the sampling or measurements;
   c) The date(s) analyses were performed;
   d) The individual(s) who performed the analyses;
   e) The analytical sampling, and sample preservation techniques or methods used;
   f) The results of such analyses.

K. Signatory Requirements: All permit applications, reports required by this permit, or other information requested by KDHE shall be signed and certified in accordance with the requirements of K.A.R. 28-46-22.

L. Reporting Requirements:

1. Except for all new wells authorized by an area permit under K.A.R. 28-46-18, a new emplacement well may not commence emplacement until construction is complete, and:

a) The permittee has submitted notice of completion of construction to KDHE; and

b) 1) KDHE has inspected or otherwise reviewed the new emplacement well and finds it is in compliance with the conditions of the permit; or

2) The permittee has not received notice from KDHE expressing the intent to inspect or otherwise review the new emplacement well within thirteen (13) days of the date of the receipt of the notice in paragraph a. of this section, in which case prior inspection or review is waived and the permittee may commence emplacement. KDHE shall allow for a reasonable time period in which the well shall be inspected.

M. Anticipated Noncompliance: If for any reason, the permittee will be unable to comply with permit requirements, the permittee shall give advance notice to KDHE. The notice shall include the reason for the anticipated noncompliance and a description of steps taken to reduce, eliminate and prevent recurrence of the noncompliance. Upon receiving proper notice from the permittee KDHE may grant for a specified time a temporary waiver to a permit requirement for the purpose of testing or treating the well or for conducting a well workover or to protect human health or the environment.

N. Transfer of Permit: This permit is not transferable to any person except after notice to and approval by KDHE. KDHE may require modification or revocation and reissuance of the permit to change the name of the permittee and incorporate such other requirements as may be necessary under the appropriate Act. The current owner shall notify KDHE at least thirty (30) days in advance of the proposed transfer date using the procedure document found on the KDHE website. The new permittee shall submit to KDHE at least thirty (30) days prior to the proposed transfer date a new permit application including the financial assurance documents guaranteeing resources are available to properly plug and abandon the well and complete facility closure.

O. Emergency Reporting: The permittee shall, within twenty-four (24) hours of becoming aware of the circumstances, orally report to KDHE any noncompliance which may endanger human health or environment. A written submission shall also be provided within five (5) days of the time the permittee becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause the period of noncompliance, including exact dates and times, corrective action taken, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance. The permittee shall comply with any corrective or remedial action required by KDHE.

P. Permit Expiration: This Permit shall be effective for a fixed term not to exceed ten (10) years. As long as KDHE is the permit-issuing authority, this Permit and all conditions herein will remain in effect beyond the Permit's expiration date per state law K.S.A. 77511 (d), which states; If a timely and sufficient application has been made for renewal of a license with reference to any activity of a continuing nature, the existing license does not expire until the state agency has taken final

action upon the application for renewal or, if the state agency's action is unfavorable, until the last day for seeking judicial review of the state agency's action or a later date fixed by the reviewing court.

Q. Severability: The provisions of this permit are severable and if any provision of this permit and any circumstance is held invalid, the application of such provision to other circumstances and the remainder of the permit shall not be affected thereby.

R. Operational Requirements:

1. The permittee shall not allow the movement of fluid containing any contaminant into any formation or aquifer not permitted to receive fluid by this permit.

2. If any water quality monitoring of an aquifer indicates the movement of any contaminant into any formation or aquifer not permitted to receive fluids by this permit, the permittee shall take such action as required by KDHE, including taking the well out of service, closure of the well or plugging and abandonment of the well.

S. Change in Emplacement Stream: Any proposed facility changes or process modifications which may result in new, different or altered Emplacement streams or an increase in emplacement volumes shall be submitted as a request to KDHE at least one hundred eighty (180) days before such changes. A permit modification may be required.

T. Permit Actions: The filing of a request by the permittee for a permit modification, a notification of planned changes or anticipated noncompliance, does not stay any permit condition.

U. Penalties: Failure to comply with the terms of this Permit may subject the Permittee to an administrative and/or civil penalty, a criminal penalty and/or an action to suspend, revoke, or terminate this Permit. Failure to minimize or mitigate any adverse impact on the environment resulting from noncompliance may serve to increase the severity of such penalties.

V. Other Conditions:

1. The permit holder shall maintain a copy of the permit and its modifications or revisions in the facility records. A copy of the permit shall also be located at the facility and accessible to those who operate the injection well.

2. Within 30 days of well recompletion, the permit holder shall submit a Permit Modification request, in accordance with K.A.R. 28-46-17, to reflect the construction details as specified in Section I and Table 1.